OSSP-PRACTICAL-WEEK2

Task1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user by accepting a Linux command as input, creating a child process using the fork() system call, executing the command in the child process using an appropriate exec() system call, allowing the parent process to wait for the child using wait(), and displaying the Process ID (PID) of both the parent and child processes.

Source code:

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    char command[100];
    char *args[20];
    char *token;
    int i = 0;

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline
    command[strcspn(command, "\n")] = '\0';

    // Split command into arguments
    token = strtok(command, " ");

    while (token != NULL && i < 19) {
        args[i++] = token;
        token = strtok(NULL, " ");
    }

    args[i] = NULL;

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID  : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
        printf("Executing command...\n");

        execvp(args[0], args);
```

```
        execvp(args[0], args);

        perror("exec failed");
        exit(1);
    }
    else {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);
        printf("Parent waiting for child...\n");

        wait(NULL);

        printf("Child process completed.\n");
        printf("Parent process completed.\n");
    }

    return 0;
}
```

Output:

```
root@LAPTOP-1COJKKG6:~# nano week2pt1.c
root@LAPTOP-1COJKKG6:~# gcc week2pt1.c -o week2pt1
root@LAPTOP-1COJKKG6:~# ./week2pt1
Enter a Linux command: ls

Parent Process
Parent PID : 520
Child PID  : 521
Parent waiting for child...

Child Process
Child PID  : 521
Parent PID : 520
Executing command: ls

week2pt1  week2pt1.c  week3pt1  week3pt1.c  week3pt2  week3pt2.c

Child process completed.
root@LAPTOP-1COJKKG6:~# |
```

Observation: I learned how the Linux operating system creates a child process using fork() and executes a Linux command using the execvp() system call. I also understood how the parent process waits for the child process using wait() and how PID and PPID help identify the relationship between parent and child processes.

Task2: Using Linux terminal commands such as uname, lscpu, lsblk, ps, and top, investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the operating system abstracts and manages CPU, memory, storage, and I/O devices.

Output:

Using command uname -a:

```
root@LAPTOP-1COJKKG6:~# uname -a
Linux LAPTOP-1COJKKG6 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
root@LAPTOP-1COJKKG6:~# |
```

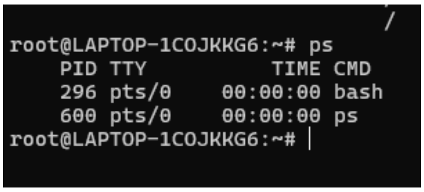
Using command lscpu:

```
root@LAPTOP-1COJKKG6:~# lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   12
  On-line CPU(s) list:    0-11
Vendor ID:                GenuineIntel
  Model name:             Intel(R) Core(TM) 5 120U
    CPU family:           6
    Model:                186
    Thread(s) per core:   2
    Core(s) per socket:   6
    Socket(s):            1
    Stepping:             3
    BogoMIPS:             4992.00
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx
                          pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq
                          vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor l
                          ahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx
                          2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip w
                          aitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    288 KiB (6 instances)
  L1i:                    192 KiB (6 instances)
  L2:                     7.5 MiB (6 instances)
  L3:                     12 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Vulnerable: No microcode
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected
root@LAPTOP-1COJKKG6:~# |
```

Using command lsblk:

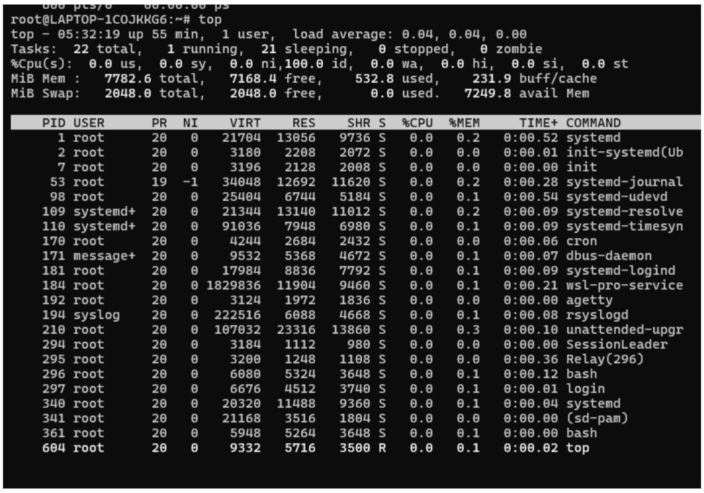
```
root@LAPTOP-1COJKKG6:~# lsblk
NAME
    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro
                              /
root@LAPTOP-1COJKKG6:~# |
```

Using command ps:



```
root@LAPTOP-1COJKKG6:~# ps
    PID TTY          TIME CMD
    296 pts/0    00:00:00 bash
    600 pts/0    00:00:00 ps
root@LAPTOP-1COJKKG6:~#
```

Using command top:



```
root@LAPTOP-1COJKKG6:~# top
top - 05:32:19 up 55 min,  1 user,  load average: 0.04, 0.04, 0.00
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7782.6 total,   7168.4 free,    532.8 used,    231.9 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7249.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   21704  13056   9736 S   0.0   0.2   0:00.52 systemd
      2 root      20   0    3180   2208   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
      7 root      20   0    3196   2128   2008 S   0.0   0.0   0:00.00 init
     53 root      19  -1   34048  12692  11620 S   0.0   0.2   0:00.28 systemd-journal
     98 root      20   0   25404   6744   5184 S   0.0   0.1   0:00.54 systemd-udevd
    109 systemd+  20   0   21344  13140  11012 S   0.0   0.2   0:00.09 systemd-resolve
    110 systemd+  20   0   91036   7948   6980 S   0.0   0.1   0:00.09 systemd-timesyn
    170 root      20   0    4244   2684   2432 S   0.0   0.0   0:00.06 cron
    171 message+  20   0    9532   5368   4672 S   0.0   0.1   0:00.07 dbus-daemon
    181 root      20   0   17984   8836   7792 S   0.0   0.1   0:00.09 systemd-logind
    184 root      20   0 1829836  11904   9460 S   0.0   0.1   0:00.21 wsl-pro-service
    192 root      20   0    3124   1972   1836 S   0.0   0.0   0:00.00 agetty
    194 syslog    20   0  222516   6088   4668 S   0.0   0.1   0:00.08 rsyslogd
    210 root      20   0  107032  23316  13860 S   0.0   0.3   0:00.10 unattended-upgr
    294 root      20   0    3184   1112    980 S   0.0   0.0   0:00.00 SessionLeader
    295 root      20   0    3200   1248   1108 S   0.0   0.0   0:00.36 Relay(296)
    296 root      20   0    6080   5324   3648 S   0.0   0.1   0:00.12 bash
    297 root      20   0    6676   4512   3740 S   0.0   0.1   0:00.01 login
    340 root      20   0   20320  11488   9360 S   0.0   0.1   0:00.04 systemd
    341 root      20   0   21168   3516   1804 S   0.0   0.0   0:00.00 (sd-pam)
    361 root      20   0    5948   5264   3648 S   0.0   0.1   0:00.00 bash
    604 root      20   0    9332   5716   3500 R   0.0   0.1   0:00.02 top
```

Observation: I learned how to use Linux commands to examine system hardware and running processes. I understood how the operating system manages CPU, storage, memory, and processes and provides these resources to applications through system services